

Intel Arc Pro B70 — Technical Reference

Xe2 “Battlemage” workstation / inference GPU · Launched 25 March 2026 · Compiled 9 August 2026

All figures below are attributed. Values marked [I] are published by Intel; [D] are derived by arithmetic from Intel-disclosed architecture rates (formula shown); [3P] are third-party measurements. Quantities Intel does not publish are stated as such rather than estimated.

1. Silicon and board

Parameter	Value	Src
Die / architecture	BMG-G31, Xe2-HPG (Battlemage)	[I]
Process	TSMC N5	[I]
Xe2 cores / render slices	32 / 8	[I]
Xe Vector Engines (XVE)	256, SIMD16 (4,096 FP32 lanes)	[I]
XMV matrix engines	256 (8 per Xe core)	[I]
Ray tracing units	32	[I]
Graphics clock / max dynamic clock	2,280 MHz / 2,800 MHz	[I]
Total board power	230 W reference; 160–290 W configurable range	[I]
Host interface	PCIe 5.0 x16	[I]
Display output	4 × DisplayPort 2.1 (UHBR 13.5); 7680×4320 @120 Hz	[I]
Media engine	2 multi-format codec engines; AV1 / HEVC / H.264 encode + decode	[I]
Reference card	10.5 x 3.9 in, dual slot, 1,020 g, 1 × 8-pin (ATX 2×4)	[I]
Warranty / launch price	3 years; \$949 list at launch	[I] / [3P]

2. Compute throughput by data type

Intel discloses two peak figures and the per-Xe-core XMV issue rates. The derived rows apply those rates at the 2,800 MHz max dynamic clock. The method reproduces Intel’s own INT8 figure exactly ($4,096 \times 32 \times 2.8 \text{ GHz} = 367 \text{ TOPS}$), which is the check on the BF16/FP16 row.

Data type	Unit	Peak throughput	Basis	Src
FP32	XVE vector	22.94 TFLOPS	Intel published, @2,800 MHz	[I]
INT8	XMV	367 TOPS	Intel published, dense	[I]
BF16 / FP16	XMV	~183.5 TFLOPS	$2,048 \text{ ops/clk/core} \times 32 \times 2.8 \text{ GHz}$	[D]
FP16	XVE vector	~45.88 TFLOPS	2:1 vs FP32	[D]
FP32	XVE vector	~18.68 TFLOPS	Same, at 2,280 MHz graphics clock	[D]

Data type	Status
TF32, INT4, INT2	Supported by the XMV engines per Intel Xe2 architecture disclosure and VTune Profiler documentation. Intel publishes no throughput figure for these.
FP64	Executes on the general-purpose XVEs, not XMV. Intel publishes no FP64 figure for this part. Third-party spec databases list ~2.87 TFLOPS on an assumed 1:8 ratio; that ratio is not confirmed by Intel and should be measured before use in any sizing model.
FP8, FP4, MXFP4	Not natively supported. Battlemage XMV acceleration extends down to FP16 and INT8 only — the principal architectural gap versus current NVIDIA parts.

3. Memory

Parameter	Value	Src
Capacity / type	32 GB GDDR6	[I]
Interface width	256-bit	[I]
Data rate	19 Gbps	[3P]
Peak bandwidth	608 GB/s	[I]

Parameter	Value	Src
ECC	Supported	[I]
vs. Arc Pro B50	Bus doubled from 128- to 256-bit; bandwidth from 224 to 608 GB/s	[3P]
Bandwidth per GB	19 GB/s per GB — decode-phase LLM inference is bandwidth-bound	[D]
Multi-GPU	4 cards = 128 GB pooled. One GPU per card requiring a full PCIe 5.0 x16 link (the B60 shipped dual-GPU at x8/x8). No NVLink-class interconnect; all inter-GPU traffic crosses PCIe.	[3P]
KV cache (Intel claim)	~93K tokens on Llama 3.1 8B BF16, vs ~42K on RTX Pro 4000 24 GB. Intel-supplied single-card Linux figure, not independently reproduced.	[I]

4. Software support

Layer	Support
Graphics APIs	DirectX 12 Ultimate, Vulkan 1.3, OpenGL 4.6, OpenCL 3.0
Compute runtime	oneAPI / SYCL, Level Zero
Inference frameworks	OpenVINO, Intel Extension for PyTorch (IPEX), PyTorch XPU backend
LLM serving	Intel LLM-Scaler — containerised vLLM fork for Battlemage under Project Battlematrix. Official B70 support landed in vllm-0.14.0-b8.2 (22 April 2026).
Fleet management	XPU Manager: firmware update, power and memory-bandwidth monitoring, diagnostics
Operating systems	Certified drivers for Windows 11, Windows 10 and Linux (xe kernel driver)
ISV certification	AutoCAD, SOLIDWORKS, Maya, Photoshop (subject to configuration requirements)
Upscaling	XeSS 2 — Super Resolution, Frame Generation, Xe Low Latency

Documented software gaps (as of June–July 2026)

- **BF16 cannot be served through vLLM.** The XPU backend serves FP16 only. Gemma 2 has no working path on the B70 as a result, despite the XMX engines supporting BF16 in silicon. Llama, Qwen, DeepSeek and Mistral families run. (Puget Systems, June 2026)
- **Quantisation coverage is incomplete.** On B60 testing, only MXFP4-native GPT-OSS models loaded; standard INT4, FP8 and AWQ failed to start. (StorageReview)
- **Container / ICD conflicts.** Intel's vllm XPU container ships Level Zero and OpenCL libraries built for Flex/Max datacentre GPUs, which collide with the B70's correct runtime. (Puget Systems)
- **Reviewer assessment.** StorageReview characterised LLM-Scaler as beta-grade, with limited model coverage and quantisation paths that do not yet work, concluding the stack — not the silicon — is the limiter.

5. Independently measured performance (selected)

Workload	Result	Configuration
Mistral Small 24B, vLLM	8,321 tok/s @ batch 32; ~65% ahead of a single RTX PRO 6000	4 × B70 (128 GB), Supermicro / EPYC 9374F, ISL 2k / OSL 256
Llama 3.1 8B, vLLM	~12,000 tok/s @ batch 32; ~85% of RTX PRO 6000	As above
GPT-OSS 120B, vLLM	6,870 tok/s @ batch 32; ~75% of RTX PRO 6000	As above
Qwen3-Coder 30B, vLLM	6,643 tok/s @ batch 32; RTX PRO 6000 led at 8,772 tok/s	As above
Stable Diffusion 1.5 FP16	Procyon 2,101 — effectively tied with RTX 5060 Ti (2,110)	Single card, Windows 11, Threadripper 7980X
Blender 4.5	1,524.6 / 846.9 / 912.2 samples per minute (Monster / Junkshop / Classroom)	Single card
Geekbench 6 OpenCL	140,165 — 2.0× the Arc Pro B50	Single card

Source: StorageReview (July 2026). Note that the multi-GPU vLLM testing used an AMD EPYC host, not Intel's all-Intel Battlematrix reference platform, which pairs these cards with a Xeon 6 host. Behaviour on Intel silicon may differ.

6. Takeaways

- **The case is VRAM capacity per dollar, not compute.** Four cards give 128 GB of ECC VRAM with workstation drivers for roughly \$3,800 at list — below a single high-end workstation GPU.
- **Throughput needs concurrency.** The card is competitive at batch 32 and unremarkable at batch 1. It suits batch and multi-user agentic serving; it does not suit single-user low-latency work.
- **The absent FP8/FP4 support is structural.** Software gaps will close with releases; the missing low-precision formats are fixed in silicon and become more limiting as FP8 KV caches and FP4 weights become standard.
- **PCIe is the scaling constraint.** With no NVLink-class fabric and x16 required per card, favour pipeline parallelism or capacity-driven sharding over aggressive tensor parallelism — and treat lane count as a first-order host-SKU filter (4 cards = 64 Gen5 lanes for GPUs alone).
- **Budget engineering time alongside capex.** Two independent reviewers hit runtime, container and precision-support problems that required manual intervention. The acquisition cost understates TCO.

Sources

- Intel — Arc Pro B70 Graphics product specifications (ark.intel.com, SKU 245797)
- Intel — Xe2 architecture disclosure (XMX per-core rates: 2,048 FP16 / 4,096 INT8 ops per clock)
- Intel VTune Profiler documentation — XMX supported data types (FP16, BF16, INT8, INT4, INT2)
- StorageReview — “Intel Arc Pro B70 Review”, 14 July 2026
- Puget Systems — “Intel Arc Pro B70 Review” (April 2026) and “Multi-GPU AI Inference Performance” (June 2026)
- Phoronix — LLM-Scaler vllm-0.14.0-b8.2 release note, 22 April 2026
- Tom's Hardware — Arc Pro B70 / B65 launch coverage, 25 March 2026
- TechPowerUp — Arc Pro B70 / B65 announcement, 25 March 2026